MATH 239 Introduction to Combinatorics

University of Waterloo - Fall 2025

12th September 2025

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Contents

1 Principles of Enumeration

Lecture 1, 2025/09/03

1.1 AND vs. OR - Products vs. Sums

Example. Let $A = \{\text{apple}, \text{orange}, \text{banana}, \text{grape}\}$ and $B = \{\text{carrot}, \text{lettuce}, \text{onion}\}$. The # of ways of choosing a fruit from A **AND** a vegetable from B is $|A| \cdot |B| = 4 \cdot 3 = 12$. This is equivalent to choosing one element from the cartesion product of A and B:

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

For finite sets, the cardinalities satisfy

$$|A \times B| = |A| \cdot |B|.$$

The # of ways of choosing a fruit from A **OR** a vegetable from B is $|A| + |B| = 4 + 3 = 7$. This is equivalent to choosing one element from the union of A and B:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

For disjoint sets,

$$|A \cup B| = |A| + |B|$$

Note that not all sets are disjoint, some have non-trivial intersection.
The intersection is $A \cap B = \{x : x \in A \text{ and } x \in B\}$. For finite sets,

$$|A \cup B| + |A \cap B| = |A| + |B|.$$

1.2 Lists, Permutations, and Subsets

Definition 1.1 (**List**).
A **list** of a set S is a listing of the elements of S exactly once, in some order.

Example. $S = \{a, b, c\}$ has 6 lists:

$$abc \quad acb \quad bac \quad bca \quad cab \quad cba.$$

We can construct a list of S by choosing an element $v \in S$ to be the first element of the list, and then append a list of $S \setminus \{v\}$. If p_n is the # of lists of a set of size n, for $n \in \mathbb{N}$, then

$$p_n = n \cdot p_{n-1}$$

and $p_1 = 1$. So by induction we have the following theorem.

Theorem 1.1. For every $n \geq 1$, the # of lists of an n-element set is $n \cdot (n-1) \cdots 2 \cdot 1 = n!$.

Note. By convention, we take $0! = 1$.

Definition 1.2 (Subset).
A **subset** of S is a set containing some (perhaps none, or perhaps all) of the elements of S, at most once each, and in no particular order.

Note. There are 2^n subsets of an n-element set.

To specify: a subset X of S, for each element $v \in S$, we have to decide whether v is in X or not in X. Equivalently, we have to choose $\{\text{in}, \text{out}\}$ n times. So,

$$\underbrace{|\{\text{in}, \text{out}\}| \cdot |\{\text{in}, \text{out}\}| \cdot \ldots \cdot |\{\text{in}, \text{out}\}|}_{n \text{ times}} = 2^n.$$

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Definition 1.3 (Partial List).
A **partial list** of a set S is a list of a subset of S.

We are interested in k-element partial lists of an n-element set, in other words, k-tuples $(s_1, \ldots, s_k)$ of distinct elements of S. There are n choices for s_1, $n-1$ choices for s_2, ..., $(n-k+1)$ choices for s_k. So we have the following.

Theorem 1.2. For $k, n \geq 0$, the # of partial lists of length k of an n-element set is

$$n(n-1)(n-2)\cdots(n-k+1).$$

Note. For $0 \leq k \leq n$, we can write

$$n(n-1)(n-2)\cdots(n-k+1) = \frac{n(n-1)\cdots 1}{(n-k)(n-k-1)\cdots 1} = \frac{n!}{(n-k)!}.$$

Definition 1.4. For $n, k \geq 0$, let $\binom{n}{k}$ (read "n choose k") denote the # of k-element subsets of an n-element set.

Example.

- $\binom{n}{0} = 1$.
- $\binom{n}{1} = n$ for all n.
- $\binom{n}{n} = 1$ for all n.
- $\binom{n}{k} = 0$ if $k > n$.

Theorem 1.3. For $0 \leq k \leq n$, we have

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Proof. Let S be a set of size n. From Theorem 1.2, the # of partial lists of S of length k is $\frac{n!}{(n-k)!}$. We will choose a k-element subset X of S AND choose a list of X, which will give a list of a subset of S of length k. Thus, the # of length-k partial lists of S is

$$(\#\text{ of } k\text{-element subsets of } S) \cdot (\#\text{ of lists of a } k\text{-element set}) = \binom{n}{k} \cdot k!.$$

So, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. □

Note. This is an example of a combinatorial proof, where we prove an identity by counting the same set in two different ways.

Example. For $n \geq 0$,

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$

Prove this by counting sets, not by algebraic manipulation.

Proof. The RHS, 2^n, is the # of subsets of an n-element set. On the LHS, for $0 \leq k \leq n$, $\binom{n}{k}$ is the # of k-element subsets of an n-element set. And addition corresponds to "OR", so the LHS is the # of ways of choosing a single subset of an n-element set. That is, choosing one of the $\binom{n}{0}$ 0-element subsets, or one of the $\binom{n}{1}$ 1-element subsets, or ..., or one of the $\binom{n}{n}$ n-element subsets. □

Example. For $k, n \geq 1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. The LHS, $\binom{n}{k}$, is the # of k-element subsets of an n-element set S. WLOG, assume $S = \{1, 2, \ldots, n\}$. Notice that $\binom{n-1}{k}$ is the # of k-element subsets of $S' = \{1, 2, \ldots, n-1\}$, and $\binom{n-1}{k-1}$ is the # of $(k-1)$-element subsets of S'. Recall our intuition that addition corresponds to "OR". Any k-element subset of S either includes n or does not include n. If $n \notin X$, then X is a subset of size k of S'. If $n \in X$, then $X \setminus \{n\}$ is a subset of size $k-1$ of S'. Thus, LHS = RHS. □

1.3 Multisets

Suppose I have a bag containing 8 marbles, each of which is either red, green, or blue. How many different bag contents are possible?

First question: how can I represent the contents of a bag? For example, we could have 2 red, 3 green, and 3 blue marbles.

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- We might want to write it as a set $\{R, R, G, G, G, B, B, B\}$, but this is not a set since sets do not have duplicates.
- Invent new notation that preserves multiplicity:

$$[[R, R, G, G, G, B, B, B]] \text{ or } << R^2, G^3, B^3 >> .$$

- Keep track of multiplicities in a vector. Fix an order, say, R then G then B, and use a tuple or a vector to record the multiplicities: $(2, 3, 3)$.

To count the # of bags containing 8 marbles each of which is either red, green, or blue, we need to count the # of 3-tuples (n_1, n_2, n_3) of integers s.t. $n_1 + n_2 + n_3 = 8$ and $n_1, n_2, n_3 \geq 0$.

Definition 1.5 (**Multiset**).
Let $n \geq 0$, $t \geq 1$ be integers. A **multiset** of size n with elements of t types is a sequence of non-negative integers $m_1, m_2, \dots, m_t$ s.t. $m_1 + m_2 + \cdots + m_t = n$. The set of all multisets of size n with elements of t-types is denoted $\mathcal{M}(n, t)$.

Theorem 1.4. For $n \geq 0$, $t \geq 1$, the # of n-element multisets with elements of t types is

$$|\mathcal{M}(n,t)| = \binom{n+t-1}{t-1}.$$

Proof. We know that $\binom{n+t-1}{t-1}$ is the # of $(t-1)$-element subsets of an $(n+t-1)$-element set. We will show how to translate between $(t-1)$-element subsets of an $(n+t-1)$-element set and multisets of size n with t types. Let us work down a row of $n+t-1$ circles (e.g. $n = 8$, $t = 3$):

$$\circ \circ \circ \circ \circ \circ \circ \circ \circ \circ$$

Now, let us choose a $(t-1)$-element subset of these and cross them out

$$\circ \circ \times \circ \circ \circ \times \circ \circ \circ$$

Our row now has n circles grouped into t segments each containing 0 or more consecutive circles. Let m_i be the length of the i^{th} segment of consecutive circles, so $m_1+m_2+\cdots+m_t = n$. So $(m_1, m_2, \dots, m_t)$ is a multiset of size n with elements of t types.

Now, let us go the other way. Let $(m_1, m_2, \dots, m_t)$ be a multiset of size n with t types. Write down m_1 circles, then an $\times$, then m_2 circles, then an $\times$, and so on, upto m_t circles. We will have written down $n + t - 1$ symbols total, of which $t - 1$ are $\times$'s, these indicate the $(t-1)$-element subset of an $(n + t - 1)$-element set. $\square$

Note. This is an example of a bijective proof.

1.4 Bijections

Definition 1.6 (**Surjective, Injective, Bijective**).

Let A and B be sets. Let $f : A \to B$.

- f is **surjective** (or **onto**) if, for every $b \in B$, $\exists a \in A$ s.t. $f(a) = b$.
- f is **injective** (or **one-to-one**) if, for every $a, a' \in A$, if $f(a) = f(a')$, then $a = a'$.
- f is **bijective** if it is both surjective and injective.

Definition 1.7 (**Well-defined**).

An expression yields a **well-defined** function $f : A \to B$ if, for every input $a \in A$, $f(a)$ assigns a single unambiguous value $b \in B$.

Proposition 1.5. A functtion $f : A \to B$ is a bijection $\iff$ f has an inverse, namely if $\exists$ a well-defined function $g : B \to A$ s.t. $g(f(a)) = a$ for all $a \in A$ and $f(g(b)) = b$ for all $b \in B$.

Remark (**Notation**).

If $\exists$ a bijection between sets A and B, we can write $A \rightleftharpoons B$.

Corollary 1.6. If $A \rightleftharpoons B$ and at least one of A or B is finite, then $|A| = |B|$.

Example. For $k, n \geq 1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. Let $S = \{1, 2, \ldots, n\}$ and $S' = \{1, 2, \ldots, n-1\}$. Let $A = \{k\text{-element subsets of } S\}$ and $B = \{(k-1)\text{-element or } k\text{-element subsets of } S'\}$. By definition, $|A| = \binom{n}{k}$ and we can easily see that $|B| = \binom{n-1}{k-1} + \binom{n-1}{k}$. The goal is to prove $A \rightleftharpoons B$. Define $f : A \to B$ by

$$f(u) = \begin{cases} u, & \text{if } n \notin u \\ u \setminus \{n\}, & \text{if } n \in u \end{cases}$$

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Question: Is f well-defined, i.e. $\forall u \in A$, is $f(u) \in B$?

Answer: Yes, $f(u)$ does not contain n (so it's a subset of S') and has size either $k-1$ or k. So

$f(u) \in B$.

Define $g : B \to A$ by

$$g(v) = \begin{cases} v, & \text{if } |v| = k \\ v \cup \{n\}, & \text{if } |v| = k - 1 \end{cases}$$

Claim (1)**.** For all $u \in A$, $g(f(u)) = u$.

Proof of Claim (1). First consider $u \in A$ s.t. $n \notin u$. Then, $f(u) = u$ and $|f(u)| = k$, so $g(f(u)) = u$ (first branch of g). Now consider $u \in A$ s.t. $n \in u$. Then, $f(u) = u \setminus \{n\}$ and $|f(u)| = k - 1$, so $g(f(u)) = (u \setminus \{n\}) \cup \{n\} = u$ (second branch of g). □

Claim (2)**.** For all $v \in B$, $f(g(v)) = v$.

Proof of Claim (2). Exercise. □

Thus, f is a bijection, so $|A| = |B|$, i.e. $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. □

Remark (**Problem Solving Tips: what steps are needed for a bijective argument?**).

To show a function $f : A \to B$ is a bijection by giving an inverse:

(1) State the function $f : A \to B$.

(2) Show that f is well-defined, i.e. $\forall a \in A$, $f(a) \in B$.

(3) State the function $g : B \to A$.

(4) Show that g is well-defined.

(5) Show that $g(f(a)) = a$ for all $a \in A$.

(6) Show that $f(g(b)) = b$ for all $b \in B$.

Exercise: Show that $\binom{n}{k} = \binom{n}{n-k}$ for $n \geq k \geq 0$ using a combinatorial or bijective argument.

1.5 Summary of Combinatorial and Bijective Proofs

In a combinatorial proof, describe two different ways of counting the size of the same set.

$$\begin{tikzcd}
& \text{Set } A \arrow[dl] \arrow[dr] & \\
\text{count using one method} \arrow[d] & & \text{count using another method} \arrow[d] \\
\text{size}_1 & = & \text{size}_2
\end{tikzcd}$$

In a bijective proof, describe two different sets and show they're in bijection.

$$\begin{tikzcd}
\text{Set } A \arrow[r, leftrightarrow, "f"] \arrow[d] & \text{Set } B \arrow[d] \\
\text{size}_A \arrow[r, phantom, "="] & \text{size}_B
\end{tikzcd}$$

(1) Give sets A, B and count them.

(2) Show a bijection between A and B.

(3) This implies $|A| = |B|$.

2 Generating Series

Definition 2.1 (**Formal Power Series**).

A **formal power series** is an expression of the form

$$G(x) = \sum_{n=0}^{\infty} g_n x^n$$

in which the coefficients $(g_0, g_1, g_2, \ldots)$ are a sequence of real numbers.

Note. We use x as an **indeterminate** for which we do not normally substitute in particular values. This means that we never worry about the convergence of the series, all we require is that the coefficients are finite real numbers.

Example. $G(x) = 0 - 1x + 2x^2 - 3x^3 + \cdots = \sum_{n=0}^{\infty} (-1)^n n x^n$.

We can add and multiply formal power series like polynomials. Let

$$F(x) = \sum_{n \geq 0} f_n x^n = f_0 + f_1 x + f_2 x^2 + \cdots,$$
$$G(x) = \sum_{n \geq 0} g_n x^n = g_0 + g_1 x + g_2 x^2 + \cdots.$$

Then,

$$(F + G)(x) = \sum_{n \geq 0} (f_n + g_n) x^n = (f_0 + g_0) + (f_1 + g_1)x + (f_2 + g_2)x^2 + \cdots,$$
$$(F \cdot G)(x) = \sum_{n \geq 0} \left(\sum_{k=0}^{n} f_k g_{n-k} \right) x^n = (f_0 g_0) + (f_0 g_1 + f_1 g_0)x + (f_0 g_2 + f_1 g_1 + f_2 g_0)x^2 + \cdots.$$

Lecture 5, 2025/09/12